

# 1 - LEP Primes

1. Assume a notation that,  $\alpha_p$  denotes the power of prime  $p$  in  $n$ .

$$\text{Now, } \gcd(m, n) = \prod_p p^{\min(\alpha_p, \beta_p)}$$

$$\begin{aligned} \text{and } \gcd(km, kn) &= \prod_p p^{\min((km)_p, (kn)_p)} \\ &= \prod_p p^{\min(k_p + \alpha_p, k_p + \beta_p)} \\ &= \prod_p p^{k_p + \min(\alpha_p, \beta_p)} \\ &= k \gcd(m, n) \quad (\text{Proved}) \end{aligned}$$

2. Assume  $n = \dots + b_2 \times p^2 + b_1 \times p^1 + b_0 \times p^0$

$$\text{Now, } \left\lfloor \frac{n}{p} \right\rfloor = \dots + b_2 \times p^1 + b_1 \times p^0$$

$$\left\lfloor \frac{n}{p^2} \right\rfloor = \dots + b_2 \times p^0$$

⋮

$$\therefore \sum_p \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \dots$$

$$= b_1 \times p^0 + b_2 \times (p^1 + p^0) + \dots$$

$$= b_1 \times \left( \frac{p-1}{p-1} \right) + b_2 \times \left( \frac{p^2-1}{p-1} \right) + \dots$$

$$\begin{aligned} &= \frac{(b_0 + b_1 \times p + b_2 \times p^2 + \dots) - (b_0 + b_1 + b_2 + \dots)}{p-1} \\ &= \frac{n - \sum_p(n)}{p-1} \quad (\text{Ans}) \end{aligned}$$

$$\begin{aligned}
3. \quad & \gcd(12n+1, 30n+2) \\
&= \gcd(12n+1, (30n+2) \bmod 12n+1) \quad [\text{Euclid}] \\
&= \gcd(12n+1, 6n) \\
&= \gcd(6n, (12n+1) \bmod 6n) \\
&= \gcd(6n, 1) \\
&= 1
\end{aligned}$$

## 2 - SB and MA

1. Assume  $n = b_0 \times 10^0 + b_1 \times 10^1 + b_2 \times 10^2 + \dots + b_k \times 10^k$

Now,  $10 \equiv 1 \pmod{3}$

$$\Rightarrow 10^n \equiv 1 \pmod{3}$$

$$\Rightarrow 10^n - 1 \equiv 0 \pmod{3}$$

$$\begin{aligned}
\therefore n &\equiv (b_0 \times 10^0 + b_1 \times 10^1 + \dots + b_k \times 10^k) \pmod{3} \\
&\equiv (b_0 + b_1 + \dots + b_k) \pmod{3}
\end{aligned}$$

$$n \equiv (\text{sum of digits}) \pmod{3}$$

$\therefore n$  is div' by 3 iff sum-digits div by 3

2. // can be done by Legendre, but maybe simpler way exists

3.  $n^i \equiv 1 \pmod{m}$

$$\therefore n^{ia} \equiv 1 \pmod{m}$$

$$n^j \equiv 1 \pmod{m}$$

$$n^{jb} \equiv 1 \pmod{m}$$

// Thanks to Sowdha

$$\therefore n^{ia} \equiv n^{jb} \pmod{m}$$

$$\therefore n^{ia} - n^{jb} \equiv 0 \pmod{m}$$

i. for any  $a, b$ ,  $n^{ia} - n^{jb} \equiv 0 \pmod{m}$

$$\therefore n^{jb} (n^{ia-jb} - 1) \equiv 0 \pmod{m}$$

$$\text{Now, } n^{jb} \equiv 1 \pmod{m}$$

$$\therefore n^{ia-jb} \equiv 1 \pmod{m}$$

Now, we can find  $a, b$  such that

$$ia - jb = \gcd(i, j) \text{ by Bezout.}$$

$$\therefore n^{\gcd(i, j)} \equiv 1 \pmod{m}$$

4.  $n(n+1)$  is surely even as one of them is even.

$$\text{Now, } 2 \mid n(n+1)(2n+1)$$

we need to prove,  $3 \mid n(n+1)(2n+1)$

if,  $n = 3k$  or,  $3k-1$  we are done,

$$\text{if } n = 3k+1,$$

$$\begin{aligned} n(n+1)(2n+1) &= n(n+1)(6k+3) \\ &= 3n(n+1)(2k+1) \end{aligned}$$

$$\therefore \text{in all cases, } 3 \mid n(n+1)(2n+1)$$

$$\therefore 6 \mid n(n+1)(2n+1) \text{ (Proved)}$$

### 3-Residue NS and FLT

1. By FLT, we know,  $a^{p-1} \equiv 1 \pmod{p}$   $\therefore a^6 \equiv 1 \pmod{7}$  if  $a \not\equiv 0 \pmod{7}$

$$\begin{aligned} \therefore 2^6 &\equiv 1 \pmod{7} \\ 2^{18} &\equiv 1 \pmod{7} \\ 2^{20} &\equiv 4 \pmod{7} \end{aligned}$$

Similarly,

$$3^{30} \equiv (3^6)^5 \equiv 1 \pmod{7}$$

$$4^{40} \equiv (4^6)^6 \cdot (4^2)^2 \equiv 4 \pmod{7}$$

$$5^{50} \equiv (5^6)^8 \cdot 5^2 \equiv 4 \pmod{7}$$

$$6^{60} \equiv (6^6)^{10} \equiv 1 \pmod{7}$$

$$\therefore \text{Ans.} = 4 + 1 + 4 + 4 + 1 \equiv 0 \pmod{7}$$

2.  $p, 8p^2+1$  are primes.  $p$  is either  $3k$  or  $3k+1$  or  $3k+2$

Now, if  $p = 3k+1$ ,  $8p^2+1$

$$= 8 \times (3k+1)^2 + 1$$

$$= 8(9k^2 + 6k + 1) + 1 = 72k^2 + 48k + 9 = 3[24k^2 + 16k + 3]$$

$3 \mid 8p^2+1$  and  $8p^2+1 > 3$  so,  $8p^2+1$  is nonprime

if  $p = 3k+2$ ,  $8p^2+1$

$$= 8(3k+2)^2 + 1$$

$$= 8(9k^2 + 12k + 4) + 1 = 72k^2 + 96k + 33$$

$$= 3(24k^2 + 32k + 11)$$

$\therefore 3 \mid 8p^2+1$  and  $8p^2+1 > 3$  so,  $8p^2+1$  is nonprime

Therefore,  $p$  must be of the form  $3k$

$$\therefore p = 3, 8p^2+1 = 73 \quad (\text{Ans})$$



# 4-ETF

1. Assume:  $n = p_1^{m_1} p_2^{m_2} p_3^{m_3} \dots p_k^{m_k}$  where,

if  $p_1 = 2$  and  $m_1 > 1$ ,

$$p_1 < p_2 < \dots < p_k$$

we would have,  $\phi(n) = (p_1^{m_1} - p_1^{m_1-1}) (\dots) (\dots)$

$$= p_1^{m_1-1} (p_1 - 1)$$

even

Thus, here  $\phi(n)$  is even

# Now, All other primes except for 2 are odd.

Say, we have a factor  $p_2^{m_2}$  which will contribute,

$$p_2^{m_2} - p_2^{m_2-1} = (p_2^{m_2-1})(p_2 - 1)$$

Since,  $p_2$  is odd,  $(p_2 - 1)$  is even.

So, if  $n$  has at least one odd prime factor, we are done.

Otherwise, if  $p_1 = 2$ ,  $m_1 > 1$ , we are done (by prev)

The only remaining case is  $n = 2$ ,

for which,  $\phi(n) = 1$ , which is not even.

$\phi(n)$  is even if  $n > 2$ .

2.  $a|b$

Now, assume,  $a = p_1^{m_1} p_2^{m_2} \dots p_k^{m_k}$

Then, we can write,

$$b = p_1^{m_1+n_1} p_2^{m_2+n_2} \dots p_k^{m_k+n_k} \left( \dots \right)$$

$$\text{therefore, } \phi(a) = a \cdot \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_k}\right)$$

$$\phi(b) = b \cdot \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_k}\right)$$

$\therefore$  when we divide  $\phi(b)$  by  $\phi(a)$ , the  
 $\phi(a)$  portions  $\left[ \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right) \right]$  will cancel out.

furthermore, the remaining part must be integer.  
 as, the primes at those denominators have a  
 positive power in  $b$ .

This proves that  $\phi(a) | \phi(b)$ .

$b = p_1^{m_1+n_1} p_2^{m_2+n_2} \dots p_k^{m_k+n_k}$